Open Knowledge for a Sustainable Future: Research, Ethics, and Wikipedia

Week 6 — CARE, ethics, and honest persuasion

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Today

1. CARE Principles for Indigenous and Community Data
2. Case Studies
3. Ethical Persuasive Writing and Statistical Honesty
4. When Results Surprise You
5. Writing for Different Audiences

Roadmap

1. CARE Principles for Indigenous and Community Data
2. Case Studies
3. Ethical Persuasive Writing and Statistical Honesty
4. When Results Surprise You
5. Writing for Different Audiences

CARE: what and why

- **FAIR** makes data easy to find and reuse
- **CARE** asks whether that openness respects the people in the data
- Developed by the Global Indigenous Data Alliance (GIDA) (Carroll et al., 2020)
  - **Collective Benefit**: who gains from sharing this data?
  - **Authority to Control**: who decides how it can be used?
  - **Responsibility**: are researchers accountable to those represented?
  - **Ethics**: how can openness coexist with respect and consent?
- FAIR without CARE is open, but not necessarily just

CARE: Collective Benefit

- Data can empower or exploit the community it describes
- Sharing should benefit the people represented, as well as the researchers
- Benefits include:
  - knowledge preservation
  - recognition
  - capacity building
  - economic return
- Example: a corpus built with a minority-language community should also serve its schools

CARE: Authority to Control

- Communities have the right to govern how their cultural data is used and shared
- Data sovereignty: Indigenous and local groups decide access, licensing and reuse
- Authority can be expressed through traditional protocols, community review, or legal agreements
- Example: before releasing ethnobotanical recordings, get the community council's approval and write down their terms of use

CARE: Responsibility

- Data stewardship is a relationship that continues after the data is collected
- Researchers keep responsibility for context, accuracy and cultural respect
- Responsibility includes acknowledging contributors, correcting misuse, and protecting the data in the long term
- Example: keep in touch with contributors; tell them how their data is reused and what you found

CARE: Ethics

- Ethics guides decisions about collection, sharing and reuse
- FAIR data can still be unethical: collected without consent, or exposing a community to harm
- Example: before publishing traditional songs or stories, ask whether public release breaks cultural norms or sacred boundaries

CARE beyond Indigenous contexts

- CARE was written for Indigenous and community-governed data, but the questions (people, consent, benefit) apply to many online communities
- Fanfiction, collaborative writing, online art:
  - **Collective Benefit**: readers and writers learn from one another
  - **Authority to Control**: authors decide where to post, and can lock or delete works
  - **Responsibility**: moderators enforce the rules and deal with harassment
  - **Ethics**: respect creators' boundaries, content warnings and attribution norms
- Not the original CARE cases, but the same four questions apply

FAIR, CARE, and sustainability

- Openness is also an environmental question
- Reusing documented data saves repeated fieldwork, duplicated storage and wasted energy
- **FAIR**: organised, shareable data is cheaper to maintain
- **CARE**: sustainable research also keeps the trust of the people it depends on
- Together they cut both the carbon and the ethical cost of digital scholarship

Towards green open research

- Every stage of the data lifecycle costs energy: storage, transfer, computation
- Good FAIR practice (clean metadata, smaller files, open formats) cuts energy use and keeps data alive longer
- CARE partnerships replace extractive one-off projects with ones that last
- Reproducible workflows and shared infrastructure avoid repeating the same computation
- Make data efficiently open: small, documented, reused, and hosted somewhere that will last

Roadmap

1. CARE Principles for Indigenous and Community Data
2. Case Studies
   - Case Study: Archive of Our Own (AO3)
   - Case Study: AI Training Data
3. Ethical Persuasive Writing and Statistical Honesty
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AO3: an open, community archive

- **Archive of Our Own (AO3)**: a non-commercial platform for fanfiction and other transformative works
- Built and run by the Organization for Transformative Works (OTW), a volunteer nonprofit
- Millions of readers and writers; the community sets the rules

Findable and Accessible

- **Findable**: user-generated tags and filters; works can be found by fandom, pairing, trope or rating
- **Accessible:** Content is free to read through an ordinary web interface — most works need no account, and each has a stable, shareable URL
- But there is *no public API*: the OTW rate-limits bulk collection and makes no exception for researchers, who scrape the site instead
- Authors can lock a work to logged-in users: a stated access rule, as FAIR asks

Interoperable and Reusable

- **Interoperable:** users tag however they like, and volunteer *tag wranglers* then link each new tag to a *canonical* tag as a synonym, so free text becomes a filterable controlled vocabulary
- That is the real interoperability story: the structure is built by the community, by hand, not imposed by a schema
- Its limit is honest too — the vocabulary lives inside AO3. There is no API and no published dump, so cross-archive comparison means scraping
- **Reusable**: transformative reuse is the point of the site
- The terms of service spell out attribution and remix norms

AO3 and the CARE Principles

- AO3 also answers the four CARE questions:
  - **Collective Benefit**: fan culture is preserved by and for fans, not a company
  - **Authority to Control**: authors keep ownership and can delete or restrict a work at any time
  - **Responsibility**: OTW publishes its policies, requires content warnings, and trains its volunteers
  - **Ethics**: consent and inclusivity are written into the rules; moderation handles harm
- Openness here is not the same as free for all: the OTW refuses bulk scraping, including for AI training

Lessons from AO3

- FAIR and CARE can coexist: open to read, but the authors stay in control
- Metadata made by the community (tag wrangling) serves both search and the culture
- Published policies and visible moderation keep people contributing
- It scales: millions of works, run by volunteers

AI training and the CARE principles

- AI systems are trained on huge text, image and audio collections, mostly gathered without asking
- The questions:
  - Who owns or controls the data used to train AI?
  - Who benefits from the commercial and social value it generates?
  - How are communities represented, or misrepresented, in the output?
- The same four CARE questions:
  - **Collective Benefit:** Who gains from AI trained on public or cultural data?
  - **Authority to Control:** Were contributors able to consent or opt out?
  - **Responsibility:** Are data origins and attributions transparent?
  - **Ethics:** Does model training cause harm or reinforce bias?

Examples and ongoing research

- Work that connects CARE to AI:
  - Carroll et al. (2020): CARE as the complement to FAIR
  - Couldry and Mejias (2019): data collection as extraction, a new colonialism
  - Bender et al. (2021): “stochastic parrots”, trained on text nobody consented to give
  - Balan et al. (2023): DECORAIT, a registry for opting creative work in or out of AI training
- What CARE asks of AI:
  - Dataset documentation (model cards, data statements)
  - Ways to consent, be credited, and share in the benefit
  - Open to read does not mean free to exploit

Generative AI: opportunities and risks

Benefits

- **Access**: information, examples and ideas on demand
- **Efficiency**: brainstorming, translation, summarising
- **Inclusion**: lower barriers for non-native writers
- **Education**: try out styles and get feedback

Concerns

- **Accuracy**: fluent, plausible, and sometimes wrong
- **Attribution**: the authors of the training data are not credited
- **Motivation**: if nobody is credited, why write or share?
- **Transparency**: training data and usage policies undisclosed
- **Sustainability**: training and running large models costs a lot of energy

Generative AI: the rule in this course

- Generative AI helps the FAIR goals (access, reuse) and hurts the CARE goals (credit, consent, responsibility)
- So, in this course:
  - **Not for writing** — and that includes proofreading. A model would rewrite your sentences into fluent, anonymous English, which is the part you are here to learn. Spell-checker yes; rewrite no
  - **Allowed for research** — explaining a concept, suggesting search terms, pointing you at literature — **if you say so**. Name the tool and version, and what you used it for
  - What a model tells you is a lead, not a source: find the original, read it, and cite that
  - **Undisclosed use counts as academic misconduct**
- Full policy on the course website

Reading, attribution, and true knowledge

- Scholarship starts with reading the sources yourself
- To cite a work you must:
  - **Read** it in full
  - **Understand** its argument and evidence
  - **Judge** how far to trust it
  - **Attribute** it so others can check you
- An AI summary skips all four:
  - You cannot tell what was left out or distorted
  - You cannot judge the evidence you did not see
  - So nothing you build on it is yours to defend
- Reading is thinking. Writing is thinking. Skip them and citation is mimicry

When the reference does not exist

- Language models produce references with the *shape* of a citation: plausible authors, a real journal, a well-formed DOI — and no paper behind it
- These are hard to catch. The authors work in the field, the journal publishes on the topic, and the title reads like something you half remember
- **Check every reference you have not read yourself**:
  - Resolve the DOI. It must reach the paper, not an error page
  - Search the exact title in Google Scholar or on the publisher's own site
  - Match author, year, volume, issue, and pages against the record you find
  - Open it, and confirm it says what you are citing it for
- The same check catches honest mistakes — people copy journal names, volumes, and page ranges wrongly too

A reference you cannot resolve is not a reference.

Roadmap

1 CARE Principles for Indigenous and Community Data

2 Case Studies

3 Ethical Persuasive Writing and Statistical Honesty

4 When Results Surprise You

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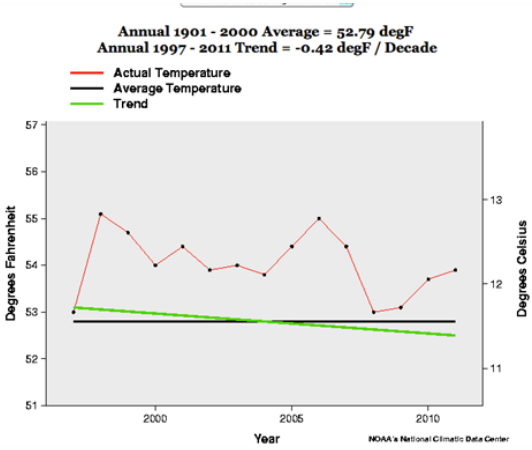
Persuasion and ethics in writing

- All academic writing persuades: we choose what to show, stress, or leave out
- Ethical persuasion presents the evidence fairly and shows the reasoning
- FAIR and CARE apply to writing too: accuracy, context, respect
- Sound data and a misleading argument is still a failure

Common pitfalls

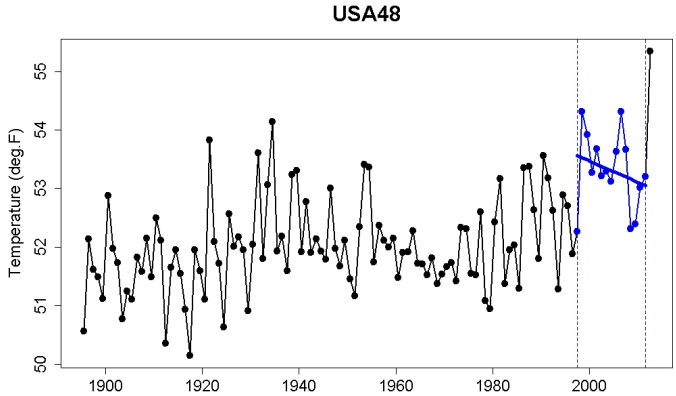
- **Cherry-picking data:** presenting only results that fit your argument
- **Overstating causality:** “X causes Y” when evidence shows only correlation
- **Ignoring confounders:** neglecting variables that explain the pattern
- **Ambiguous visuals:** missing labels, unclear units, or distorted scales
- **Rhetorical exaggeration:** emotional or absolute language (*proves, always, never*)

Cherry-picking data: The temperature is declining!



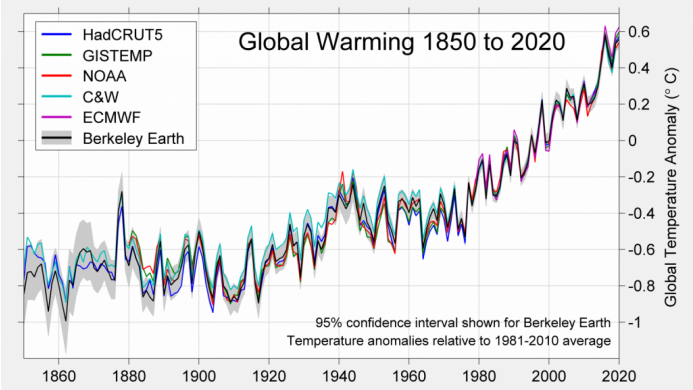
John Coleman, cited by Anthony Watts, cited by Open Mind

Not if you zoom out



Open Mind *Cherry Picking is Child's Play*

And average over different regions



Climate Change Commission *CCC: No more time for inaction as global warming accelerates, marking 2020 as one of the warmest years on record* (graph: Berkeley Earth)

Example: cherry-picking data

What was done wrong?

- Only the years supporting the argument were shown
- Broader data contradicts the claimed trend
- No context or uncertainty provided

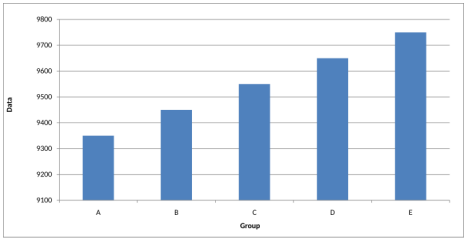
Better practice:

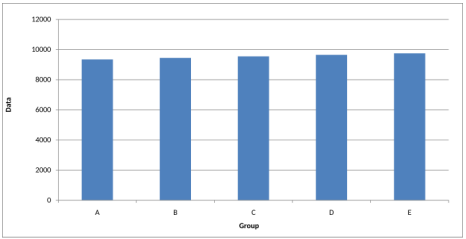
- Show full dataset and acknowledge anomalies
- Explain the anomalies
- Show confidence intervals (the grey band in the last image)

If someone cherry picks data, and refuses to acknowledge it after it has been pointed out, then they are not a credible source…

How (not) to lie with graphs

- **Truncated axes** exaggerate small changes
- **Omitted error bars** hide uncertainty
- **3D charts** distort visual proportions
- **Unequal scales** misrepresent relationships





Images from Wikipedia: Misleading graph (CC BY-SA; accessed 2026-09-20)

From data to argument

- **FAIR failures** (missing metadata, unclear methods) give weak or unverifiable claims
- **CARE failures** (ignoring context or consent) give distorted or harmful stories
- So before you cite:
  - Check where the data came from and how good it is
  - Say how sure you are
  - Describe people and communities the way they would recognise
- This is as true for citing writing as it is for citing data!

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When what you find surprises you

- You hypothesised X, and the data support Y, or nothing at all
- That is how discovery happens
- Many good findings began as “wrong” results someone looked into instead of hiding
- Report what you found, not what you hoped to find

If you discover that what you thought was wrong, *then you have improved your model of the world*; if you just confirm what you thought, you have gained less!

Ethical responses to unexpected results

- **Reflect**: why might this have happened? Re-check assumptions, methods, definitions
- **Consider alternatives**: sampling, measurement, confounders?
- **Report**: write down what changed in methods, data or interpretation
- **Revise**: update the theory to fit the evidence, not the other way round

FAIR + CARE: supporting honest revision

- **FAIR** (metadata, versioning, provenance) lets others trace how a result emerged
- **CARE** (context, consent, responsibility) says how to interpret a surprise about people
- Complete records let you, and others:
  - Reanalyse the same dataset later
  - Identify methodological errors
  - Reuse data collected by others, with its context
- This applies to found data too: when you reuse an open dataset, respect its limits and write down your reasoning

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   - Ethics in Fiction Writing

Scientific, creative, and encyclopedic writing

- **Scientific**: formal, objective, structured
  - Evidence-based, peer-reviewed, reproducible
  - Goal: to inform and test ideas transparently
- **Creative**: expressive, narrative, open to emotion and metaphor
  - Goal: to move, provoke, or entertain
  - Its truth is emotional or imaginative
- **Wikipedia**: concise, neutral, verifiable
  - Summarises the consensus; takes no side
  - Accessible to a broad, multilingual audience

Ethical imagination

- Fiction is free to invent; does that freedom come with responsibility?
- Writers shape how readers see the world, other cultures, and themselves
- **Ethical imagination**: invent without exploiting other people's pain or identity
- Ask:
  - Whose stories am I telling?
  - Do I have the right, or the context, to tell them?
  - Who might be harmed or misrepresented by this portrayal?
- Careless storytelling reinforces stereotypes; careful storytelling widens what readers can imagine

Sources: (Haynes, 2021; Khan, 2021; LeClerc, 2024)

Cultural sensitivity and responsibility

- **Cultural sensitivity** is awareness, and nobody is asking you to censor yourself
- Portray groups and traditions accurately, based on research and listening
- Avoid:
  - Cultural elements as exotic decoration (appropriation)
  - Reducing people to symbols or stereotypes
  - Borrowing trauma without accountability
- When writing outside your own experience:
  - Consult people from that background
  - Be open to feedback and revision

Sources: (Khan, 2021; Maguire, 2009; O'Callaghan, 2025)

I am not an expert here

Acknowledgements I

- **Temperature graphs:** the two US 48-state plots come from Tamino's Open Mind post *Cherry-Picking is Child's Play* (2013-03-04), which is rebutting John Coleman, reposted by Anthony Watts. The six-dataset comparison is from the Philippine Climate Change Commission, who credit Berkeley Earth.
- **Bar charts:** the truncated and untruncated bar graphs are from Wikipedia: Misleading graph (CC BY-SA).
- **CARE material** draws on the Global Indigenous Data Alliance (Carroll et al., 2020).
- **The AO3 case study** rests on the Archive of Our Own and the Organization for Transformative Works, including their statement on AI and data scraping on the Archive.
- **AI disclosure: GPT-4o** in ChatGPT (OpenAI, 2025) was used to format the references and to draft an early version of these slides.
- **Claude Opus 5** (Anthropic, 2026) was used to check the slides: to resolve every DOI against Crossref, fetch every URL, and compare each quotation with its original. The prompt was of the form

Acknowledgements II

```
Verify every claim, DOI, URL and quotation against a real source. Where you cannot
verify something, say so rather than guessing --- never invent a reference.
```

Each correction names the source it was checked against. Responsibility for what remains is mine.

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